PRODUCT REQUIREMENT DOCUMENT (PRD)

UM HACKATHON

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Project Name: Misteak– AI-Powered F&B Decision Intelligence

1.0 Project Overview

- **Problem Statement**

  Decision-making in the Food and Beverage (F&B) industry still relies heavily on intuition because there are no easily accessible and structured analytics tools. Most of the data that restaurant owners and managers deal with are fragmented data such as sales information, customer feedback and operational data but these data are seldom integrated and analyzed in a systematic manner. As a result, key business questions such as "Why revenue is stagnating, whether pricing is effective or How customer engagement translates into demand" are difficult to answer. Non-technical users, especially small business owners and entrepreneurs often lack the analytical skills, time or resources required to extract meaningful insights from raw data. Existing business intelligence tools are either too complex, require specialized expertise or are financially inaccessible for SMEs. This introduces a huge gap between data availability and decision-making capability, which results in inefficient operations, ineffective pricing and demand forecasting and finally, lost revenue opportunities.

- **Target Domain**

  Misteak is a platform of decision intelligence based on AI and serving small and medium-sized enterprises (SMEs) in the Food and Beverage (F&B) industry. The site is specifically aimed at restaurant analytics, operational optimization and strategic decision support. Misteak contrasts with the traditional analytics systems, which are primarily concentrated on dashboards and raw data visualization because it is a decision-centric system with more emphasis on interpretation and action. It aims to bridge the gap between data and decision-making by enabling users to quickly understand business performance without requiring technical expertise in data analysis. The system is well-suited for early-stage and growing businesses that need fast, reliable insights but lack dedicated data teams. Misteak positions itself as a lightweight yet powerful alternative to enterprise-grade analytics tools, enabling SMEs to leverage data-driven strategies in a practical and scalable manner.

- **Proposed Solution Summary**

  Misteak is an AI-powered decision intelligence platform that transform raw restaurant data into structured insights and actionable recommendations. The system begins by preprocessing user-uploaded datasets, including cleaning, aggregation and feature engineering to derive meaningful signals. These features are then processed through modular scoring engines such as customer score, pricing score, supply score and business score which convert different aspects of performance into standardized and interpretable metrics.

  Instead of sending raw data directly to the AI, Misteak constructs a structured JSON payload containing scores, key metrics, derived features and intermediate insights. This payload is passed to the Z.AI GLM which interprets the data, explains performance drivers and generates strategic recommendations. By combining rule-based scoring with AI reasoning, the system ensures consistent and context-aware outputs, enabling users to move from manual data analysis to faster, more informed decision making.

2.0 Background & Business Objective

- **Background of the Problem**

  The current challenges in decision-making within the F&B industry stem from the way data has historically been managed and utilized. Restaurants typically operate with multiple disconnected data sources, including sales records, customer feedback and operational data which are rarely integrated into a unified system. Over time, business owners have adapted by relying on spreadsheets or manual methods to analyze this information, but these approaches remain fragmented and limited in analytical depth. As a result, understanding the relationships between key business factors such as how customer engagement affects demand or how pricing impacts revenue become difficult. This has led to a continued dependence on intuition rather than structured, data-driven analysis

  In addition, SMEs in the F&B sector face significant constraints in adopting advanced analytics solutions. Unlike large organizations with dedicated data teams and infrastructure, smaller businesses often lack the technical expertise, time and financial resources required to implement sophisticated tools. Existing business intelligence platforms are frequently too complex or costly, making them inaccessible to many users. Consequently, there is a growing gap between the availability of data and the ability to effectively use it for decision-making. As data volume and complexity increase, traditional methods become increasingly insufficient, reinforcing the need for a system that can transform fragmented data into structured, interpretable and actionable insights.

- **Importance of Solving This Issue**

  There is a need to address this issue to enhance the efficiency and competitiveness of F&B businesses. The time taken to extract insights can be decreased by hours or days to almost real-time by converting complex data analysis into an automated and structured process. This will help owners of the business to fasten detect performance problems, experiment with strategies and adjust to the market changes. Additionally, simplifying analytics removes the technical barrier that traditionally limits data usage to skilled professionals. Through an AI-based application, Misteak, non-technical users will be able to obtain valuable insights without having to be knowledgeable in data science or analytical software. This democratization of data enables small busines.ses make smarter decisions, maximize pricing and operations and enhance customer experience. Finally, solving this issue allows businesses to move from reactive decision-making to proactive and strategic planning, leading to better financial outcomes and sustainable growth.

- **Strategic Fit / Impact**

  Misteak aligns with the broader shift toward AI-driven decision intelligence and the increasing adoption of data-centric tools across industries. By integrating structured feature engineering, modular scoring systems and the reasoning capabilities of Z.AI ensure reliability and contextual comprehension, which makes it suitable for real-world decision-making. Misteak also contributes to the digital transformation of SMEs strategically, offering an easy-to-use and scalable solution that facilitates the transition between raw data and actionable insights. It places itself in the new ecosystem of AI-native business tools where processing, analysis and decision support become an inseparable part of single workflow. In the long term, this approach enables the platform to evolve into a scalable product with potential beyond the F&B sector into other data-driven industries.

3.0 Product Purpose

3.1 Main Goal of the System

The main goal of Misteak is to serve as an AI-enabled decision intelligence partner for F&B businesses. The system is designed to transform raw, fragmented restaurant data into structured insights, performance scores and actionable recommendations that support strategic decision-making. By combining feature engineering, modular scoring systems and the reasoning capabilities of the Z.AI General Language Model (GLM), Misteak provides users with a clear understanding of their business performance, including what is happening, why it is happening and what actions should be taken. Rather than requiring users to manually analyze data or interpret complex dashboards, the platform simplifies the process by delivering concise and context-aware insights. Finally, the goal is to shift the user's role from data interpreter to decision-maker, enabling faster, more confident and data-driven business decisions.

3.2 Intended Users (Target Audience)

Misteak is designed to serve a diverse range of users within the F&B industry, especially those involved in decision-making and operational management. The primary users include:

- **Restaurant owners and managers** who require clear and timely insights to monitor business performance, optimize operations and improve profitability.

- **F&B entrepreneurs**, especially those in early-stage ventures, can leverage the system to better understand market dynamics, evaluate strategies and make data-driven decisions without needing advanced analytical skills.

- **Entry-level business analysts** who may lack experience or access to sophisticated tools. Misteak simplifies the analytical process by providing structured insights and explanations, allowing them to focus on interpretation rather than data preparation.

- **Operations managers** who benefit from the system by gaining visibility into key performance drivers, enabling them to identify inefficiencies, improve service quality and manage resources more effectively.

- **SME decision makers** who require a practical and accessible solution to bridge the gap between raw data and actionable business intelligence, supporting faster and more informed strategic decisions.

4.0 System Functionalities

4.1 Description

Misteak operates as a data-driven analytics and decision intelligence platform built using Streamlit. The system is designed to process structured restaurant data, apply feature engineering and scoring algorithms and deliver actionable insights through an integrated interface. Rathen than functioning as a traditional dashboard-only tool, Misteak combines data processing, analytical scoring and AI reasoning into a unifies pipeline. Processed data is transformed into structured performance signals, which are then interpreted and presented through dashboards, alerts, simulations and AI-generated recommendations. This enables users to move beyond simple data visualization toward understanding the underlying drivers of business performance and making informed decisions.

4.2 Key Functionalities

Misteak provides several core functionalities that work together to support end-to-end decision-making.

1. **Dashboard Analytics**

   The Dashboard Analytics module offers a comprehensive overview of business performance by displaying key performance indicators (KPIs) such as revenue, profit and trends. It visualizes sales data in an intuitive format, allowing users to quickly assess the current state of their business and identify patterns over time.

2. **Smart Alerts System**

   The Smart Alerts System enhances proactive decision-making by automatically detecting anomalies in the data, such as sudden drops in sales or unusual fluctuations in performance. These alerts notify users of potential risks or emerging opportunities, enabling timely intervention before issues escalate.

3. **Strategy Recommendations**

   The Strategy Recommendations module provides data-driven suggestions based on identified patterns and performance signals. By analyzing structured inputs such as customer engagement and operational metrics, the system recommends strategies to optimize pricing, improve product offerings and enhance overall business performance.

4. **Simulation Engine**

   The Simulation Engine allows users to evaluate hypothetical scenarios through "what-if" analysis. For example, users can simulate the impact of pricing adjustments on revenue or test different operational strategies. This functionality support strategic planning by enabling users to explore potential outcomes before implementing changes in real-world operations.

5. **AI Decision Support**

   The AI Decision Support component serves as the core intelligence layer of the system. It aggregates insights from multiple modules, including scoring engines and derived features, and uses AI reasoning to generate comprehensive explanation and recommendations. This transforms fragmented data signals into coherent, actionable guidance for decision-making.

6. **Pricing Score Engine**

   The Pricing Score Engine represents one of the system's key analytical modules. It calculates pricing effectiveness using a dedicated algorithm and outputs a structured evaluation in JSON format. This structured output is designed to be directly interpretable by the Z.AI, ensuring consistency and reliability in downstream analysis.

4.3 AI Model & Prompt Design

4.3.1 Model Selection

Misteak utilizes the Z.AI GLM as its primary AI engine for decision support and recommendation generation. The model is selected for its strong reasoning capabilities, especially in interpreting structured business data and generating context-aware insights. It is capable of handling both structure inputs (such as JSON payloads containing scores and metrics) and unstructured reasoning tasks, making it well-suited for transforming analytical signals into actionable recommendations. Additionally, the model is cost-efficient for repeated inference which is important for scalable deployment and is effective in producing clear, business-oriented outputs.

4.3.2 Prompting Strategy

The system adopts a hybrid prompting strategy that combines rule-based logic with AI-driven reasoning. Few-shot prompting is used to ensure consistency in the format and structure of the generated recommendations, guiding the model to produce predictable and interpretable outputs. At the same time, rule-based augmentation is applied through the scoring engine and derived features which provide the model with structured context such as performance scores, key metrics and intermediate insights.

This hybrid approach improves reliability by reducing the likelihood of hallucinations and ensuring that the AI operates within a well-defined analytical framework. By combining deterministic logic with flexible reasoning, the system achieves both consistency and contextual depth in its outputs.

4.3.3 Context & Input Handling

The system is designed to optimize the quality and efficiency of AI inputs. Instead of passing raw datasets directly to the model, Misteak preprocesses and aggregates the data into a structured format before AI interaction. The primary input is a processed CSV dataset which undergoes cleaning, normalization and feature engineering. From this, a structured JSON payload is generated containing performance scores, key metrics, derived signals and insights.

To handle large datasets, the system applies aggregation techniques to reduce input size while preserving essential information. If necessary, data can be chunked or summarized to fit within model constraints. Extremely large or incompatible datasets are either rejected or simplified to ensure stable system performance. This approach ensures that the AI model receives high-quality, context-rich outputs without exceeding computational limits.

4.3.4 Fallback & Failure Behavior

Misteak incorporates robust fallback mechanisms to ensure reliability and user trust. If the AI generates off-topic or irrelevant outputs, the system automatically retries with a refined prompt to improve alignment. In cases where the output remains unusable, the system falls back to rule-based insights derived from the scoring engine, ensuring that users still receive meaningful guidance. To mitigate hallucination risks, AI outputs are cross-checked against structured data such as pricing scores and derived metrics.

Additional safeguards include clear and graceful error messages within the user interface, ensuring that users are informed of any issues without confusion. The system avoids silent failures by always providing feedback or alternative outputs. Overall, the integration of deterministic logic with AI reasoning ensures a reliable and consistent user experience, even in edge cases or failure scenarios.

5.0 User Stories & Use Cases

5.1 User Stories

- As a restaurant owner, I want to upload my business data so that I can understand my overall performance without manual analysis.
- As a manager, I want to receive alerts on unusual trends so that I can react quickly to potential issues or opportunities.
- As a decision maker, I want AI recommendations so that I can improve pricing, operations and business strategies.
- As a user, I want to simulate different scenarios so that I can evaluate the impact of decisions before applying them.
- As a non-technical user, I want insights presented in a simple and structured way so that I can make decisions without needing data expertise.

5.2 Use Cases (Main Interactions)

Prompt to UI Flow:

Users interact with Misteak by uploading their restaurant dataset or providing input through the interface. The system first preprocesses the data by cleaning, handling missing values and removing inconsistencies. It then performs data aggregation and feature engineering to derive meaningful signals such as customer engagement levels and experience quality gaps. These processed features are passed into the scoring engine, which generates standardized performance scores across different business dimensions

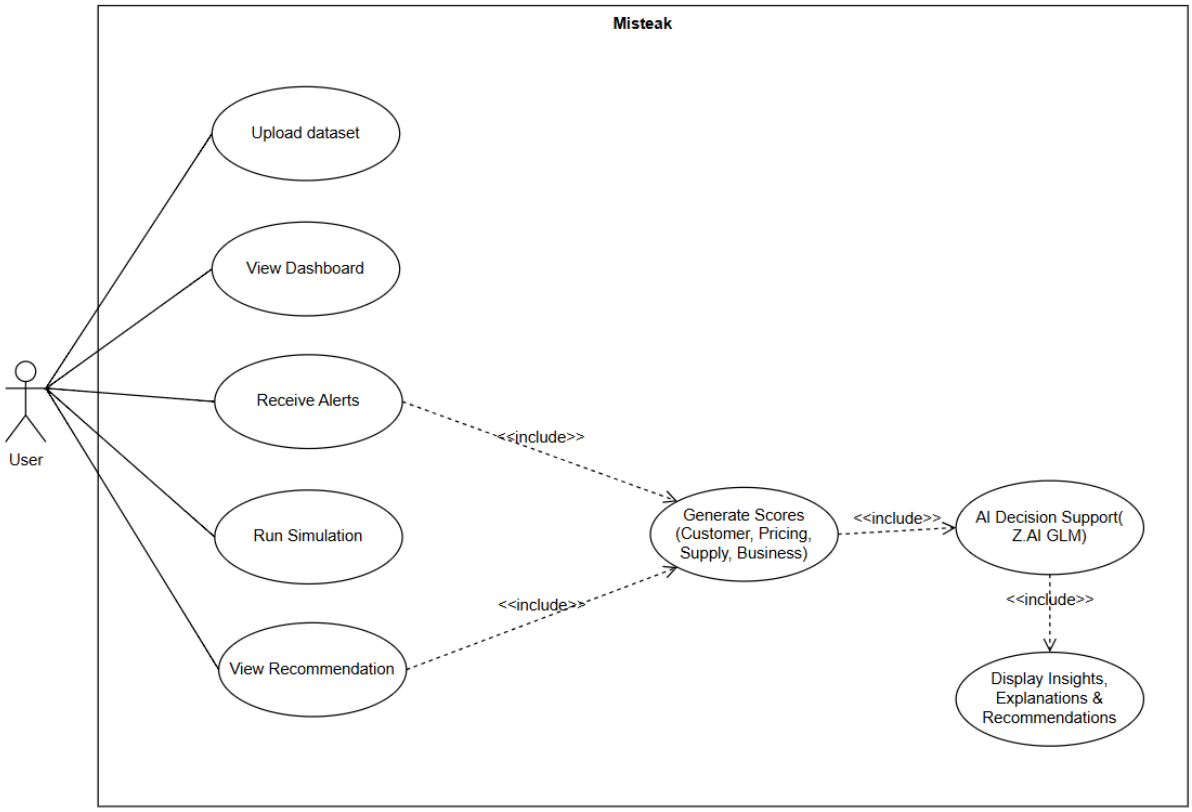
The system then constructs a structured JSON payload containing scores, key metrics, derived features and intermediate insights. This payload is sent to the Z.AI GLM which interprets the data, explains the underlying drivers of performance and generates recommendations. Finally, the results are rendered in the user interface through dashboards, alerts and recommendation panels, allowing users to easily understand and act on the insights without needing to interpret raw data.

Flow Generation:

When users interact with specific features of the system such as simulation or strategy recommendations, Misteak dynamically generates outputs based on selected function. For example, when a user inputs a scenario in the simulation module, the system recalculates relevant metrics and scores, updates the structured payload and sends it to the Z.AI model for interpretation. The model then produces updated insights and projected outcomes which are displayed in the interface.

Similarly, when users access different modules such as the dashboard or alerts system, the platform retrieves relevant data, applies scoring logic and updates the visualizations and insights in real time. Throughout this process, the system maintains consistency in scoring, data representation and output format, ensuring a seamless and coherent user experience across all functionalities.

Use Case Diagram:



6.0 Features Included (Scope Definition)

- **Interactive dashboard (Streamlit UI)**
  Provides a user-friendly interface to visualize key business metrics, trends and overall performance.
- **KPI monitoring**
  Tracks important performance indicators such as revenue, profit and operational metric as data is processed.
- **Alert detection system**
  Automatically detects anomalies (such as sudden drop in sales) and notifies users of potential risks or opportunities.
- **Strategy recommendation module**
  Generates data-driven recommendations based on scoring outputs and AI analysis to improve business performance.
- **Simulation engine**
  Enables users to perform "what-if" analysis by testing different scenarios (such as pricing changes) and evaluating potential outcomes.
- **Pricing score calculation**
  Applies algorithmic logic to evaluate pricing effectiveness and outputs structured results for further analysis.
- **AI-based decision support (Z.AI GLM)**
  Interprets structured data, explains performance drivers and generates actionable insights and recommendations.
- **Multi-page navigation system**
  Allows users to seamlessly navigate between different modules such as dashboard, alerts, simulation and recommendations.

7.0 Features Not Included (Scope Control)

- **Real-time database integration**
  The system currently operates on static CSV files and does not support live data streaming or database connections.
- **Multi-user collaboration**
  The platform does not include features for multiple users to collaborate, share access or manage roles within the system.
- **Mobile application version**
  Misteak is designed as a web-based application and does not provide a dedicated mobile application.
- **Backend API deployment**
  The system is not deployed as a scalable backend service and does not expose APIs for external integration.
- **Integration with POS systems**
  There is no direct integration with POS system or third-party restaurant management tools.
- **Production-grade scalability**
  The current implementation is a prototype and is not optimized for high concurrency large-scale data processing or enterprise-level deployment.

8.0 Assumptions & Constraints

LLM Cost Constraint

- Estimated Cost per Session
- The cost per user session is relatively low due to the use of summarized and structured inputs instead of raw datasets.
- Cost Optimization Strategies
- Data is preprocessed and aggregated before sent to the Z.AI, reducing token usage.
- Structured JSON payloads are used instead of raw data to minimize input size.
- Repeated queries can be cached to avoid redundant API calls.
- Prompt size is controlled to ensure efficient inference and cost management is scale.

Technical Constraints

- The system is built using Streamlit which is suitable for prototyping but not optimized for enterprise-scale deployment.
- The implementation is limited to a Python-based environment, restricting compatibility with other technology stacks.
- The platform does not support real-time data streaming and relies on static datasets.

Performance Constraints

- Large datasets may increase processing time due to preprocessing, aggregation and scoring computations.
- AI response quality and speed depend heavily on prompt design and input structure.
- There are practical limits on input size and processing capacity to ensure stable system performance.

User Input Constraints

- The system requires structure or semi-structured data to function effectively.
- The AI model cannot operate without user-provided data or inputs.
- Human validation is still required to verify AI-generated insights and recommendations, especially in critical decision-making scenarios.

9.0 Risks & Questions Throughout Development

- **Insight Consistency**

  How do we ensure that the Z.AI generates consistent insights across different users and datasets, instead of producing generic or repetitive recommendations?

- **System Stability**

  What happens if the data processing pipeline fails? Does the system handle failures gracefully or does it break the workflow?

- **Data Interpretation**

  How does the system handle cases where the input dataset lacks sufficient context or contains ambiguous signals? Can the AI still

- **Input Variability**

  How can the system adapt to different dataset structures uploaded by users while maintaining consistent scoring logic and output format?

- **Output Reliability**

  How do we ensure that the AI explanations are grounded in actual data signals rather than hallucinated or overly generalized responses?